

GEORGE ISLAND, MB, Canada

WMO: 714450

Lat: 52.819N

Lon: 97.616W

Elev: 730

StdP: 14.31

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.1	-21.7	-31.5	1.0	-24.9	-27.7	1.2	-21.4	35.4	-2.3	32.0	-2.1	10.8	320	0.822

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.8	75.5	67.4	73.3	66.4	71.5	65.4	69.5	73.3	67.8	71.8	66.3	70.5	9.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.0	105.8	72.1	66.2	99.0	70.7	64.5	93.3	69.3	34.1	73.5	32.7	72.0	31.4	70.6	76.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
35.0	30.4	27.2	-29.7	80.1	5.6	2.0	-33.7	81.5	-37.0	82.7	-40.2	83.8	-44.3	85.2
			-29.9	71.2	5.4	1.7	-33.8	72.4	-36.9	73.3	-39.9	74.2	-43.8	75.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	33.5	-0.9	1.7	14.3	27.7	40.4	55.7	65.2	64.9	56.4	41.9	25.4	7.3
	DBStd	25.16	13.50	12.51	13.42	9.86	7.00	6.89	4.73	4.78	6.73	7.00	10.93	12.86
	HDD50	7383	1578	1354	1108	668	304	25	0	0	21	264	738	1323
	HDD65	11630	2043	1774	1573	1118	761	287	55	61	266	716	1188	1788
	CDD50	1362	0	0	0	0	8	194	473	462	212	13	0	0
	CDD65	135	0	0	0	0	0	7	63	58	7	0	0	0
	CDH74	129	0	0	0	0	0	8	71	49	1	0	0	0
	CDH80	2	0	0	0	0	0	0	2	0	0	0	0	0
(14) Wind	WSAvg	13.4	13.1	12.6	12.6	12.3	12.7	11.5	11.7	13.5	15.1	16.2	15.7	13.6
(15) Precipitation	PrecAvg	18.3	0.7	0.4	0.8	0.7	1.7	2.6	2.2	2.4	2.5	1.5	1.1	0.7
	PrecMax	24.1	2.4	0.8	4.0	1.7	3.4	5.9	5.4	6.1	5.3	5.2	3.6	1.8
	PrecMin	7.2	0.0	0.1	0.0	0.0	0.0	0.7	0.1	0.0	0.4	0.1	0.0	0.0
	PrecStd	3.7	0.6	0.2	0.9	0.5	0.9	1.3	1.4	1.6	1.2	1.2	0.8	0.5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.9	34.8	41.7	49.0	61.9	74.7	78.7	77.8	71.8	59.2	46.8	35.5
		MCWB	32.7	32.4	36.8	41.3	51.6	64.5	68.5	68.5	64.2	53.7	43.6	33.5
	2%	DB	28.4	30.9	38.0	44.5	55.7	70.6	76.0	75.3	69.2	56.6	43.5	31.5
		MCWB	27.1	29.4	34.4	38.4	48.1	62.6	67.7	67.5	63.2	52.2	40.2	30.1
	5%	DB	23.9	26.2	35.7	41.4	52.3	67.6	73.8	73.3	67.3	54.2	41.4	28.5
		MCWB	22.9	24.8	33.1	36.8	46.5	60.7	66.8	66.7	61.8	50.1	38.5	27.4
	10%	DB	19.4	20.6	33.5	39.0	49.6	65.1	72.0	71.4	65.2	52.0	39.2	25.1
		MCWB	18.6	19.5	31.7	35.4	44.9	59.5	65.8	65.7	60.0	48.3	36.6	24.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	33.1	32.4	37.3	42.2	53.1	66.3	72.3	71.5	66.4	56.0	44.1	33.4
		MCDB	34.1	33.8	40.7	47.8	58.9	71.9	75.9	74.7	70.3	58.5	46.5	35.0
	2%	WB	27.2	28.6	34.8	39.2	49.5	64.2	69.8	69.7	64.2	52.9	40.9	30.1
		MCDB	28.3	30.0	37.3	43.4	54.3	69.2	73.9	73.2	67.9	55.7	43.0	31.1
	5%	WB	22.9	24.1	33.2	37.2	47.3	62.3	68.1	68.1	62.5	50.5	38.6	27.3
		MCDB	23.9	25.5	35.3	40.7	51.6	66.4	72.2	71.9	66.6	53.7	41.0	28.4
	10%	WB	18.4	18.9	31.8	35.7	45.3	60.2	66.6	66.5	60.7	48.4	36.8	23.8
		MCDB	19.2	19.9	33.9	38.8	49.2	64.5	70.9	70.6	65.0	51.7	39.2	24.9
(35) Mean Daily Temperature Range	MDBR		13.2	15.4	16.8	14.3	11.3	10.5	8.8	8.0	7.9	6.8	7.8	10.7
	5% DB	MCDBR	15.4	17.4	14.9	14.5	15.6	13.3	10.6	9.5	9.0	8.4	6.5	11.0
		MCWBR	15.0	17.4	12.4	9.8	9.2	7.2	6.2	5.9	7.1	6.4	5.6	10.3
	5% WB	MCDBR	15.6	18.4	13.9	13.2	14.1	11.4	9.7	8.9	8.5	7.6	6.4	11.0
		MCWBR	15.2	17.4	11.6	9.4	9.1	7.1	6.5	6.2	7.1	6.4	5.8	10.4
(40) Clear-Sky Solar Irradiance	taub		0.255	0.254	0.308	0.346	0.354	0.371	0.387	0.363	0.329	0.295	0.252	0.232
	taud		2.211	2.280	2.253	2.358	2.429	2.394	2.375	2.472	2.526	2.567	2.363	2.255
	Ebn at Noon		233	277	279	282	285	279	272	273	272	262	239	225
	Edh at Noon		20	26	33	35	34	36	36	31	26	20	17	16
(44) All-Sky Solar Radiation	RadAvg		276	590	1003	1489	1742	1819	1799	1486	1031	558	281	203
	RadStd		28	46	94	128	133	130	93	108	111	62	29	28

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	N/A	N/A	+3.40	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page